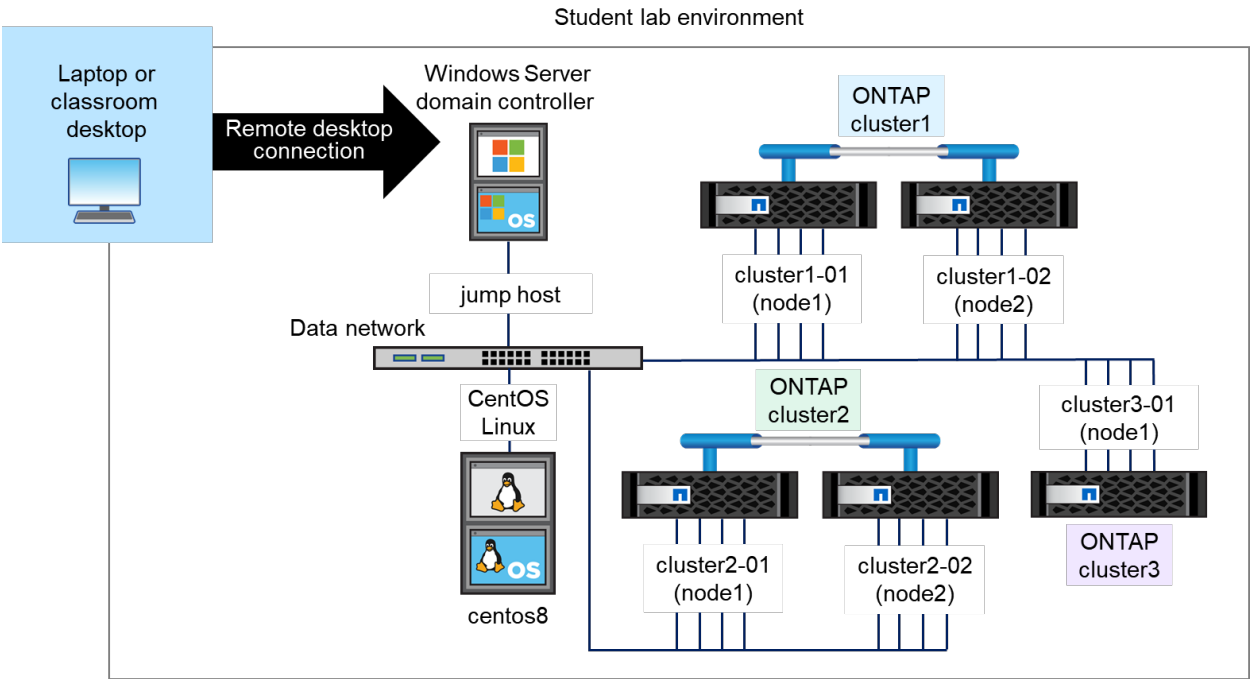


Module 4: Storage VM Disaster Recovery (SVM DR)

Exercise Equipment Diagram



System	Host Name	IP Addresses	User Name	Password
Windows Server 2022 Datacenter	JUMPHOST	192.168.0.5	DEMO\Administrator	Netapp1!
ONTAP cluster-management LIF (Cluster1)	Cluster1	192.168.0.101	admin (case-sensitive)	Netapp1!
node 1 (Cluster1)	cluster1-01-console	192.168.0.111	admin (case-sensitive)	Netapp1!
node 2 (Cluster1)	cluster1-02-console	192.168.0.112	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster2)	Cluster2	192.168.0.102	admin (case-sensitive)	Netapp1!
node 1 (Cluster2)	cluster2-01-console	192.168.0.113	admin (case-sensitive)	Netapp1!
node 2 (Cluster2)	cluster2-02-console	192.168.0.114	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster3)	Cluster3	192.168.0.103	admin (case-sensitive)	Netapp1!
node 1 (Cluster3)	cluster3-01-console	192.168.0.115	admin (case-sensitive)	Netapp1!
Linux Server	centos8	192.168.0.21	root	Netapp1!

Exercise 1: Configuring Storage VM DR

In this exercise, you use NetApp ONTAP System Manager to configure a disaster recovery (DR) relationship between two storage VMs (storage virtual machines, also known as SVMs) in different clusters. You can also use the ONTAP CLI to configure SVM DR. You use the c1_svm3 storage VM as the source storage VM.

Objectives

This exercise focuses on enabling you to do the following:

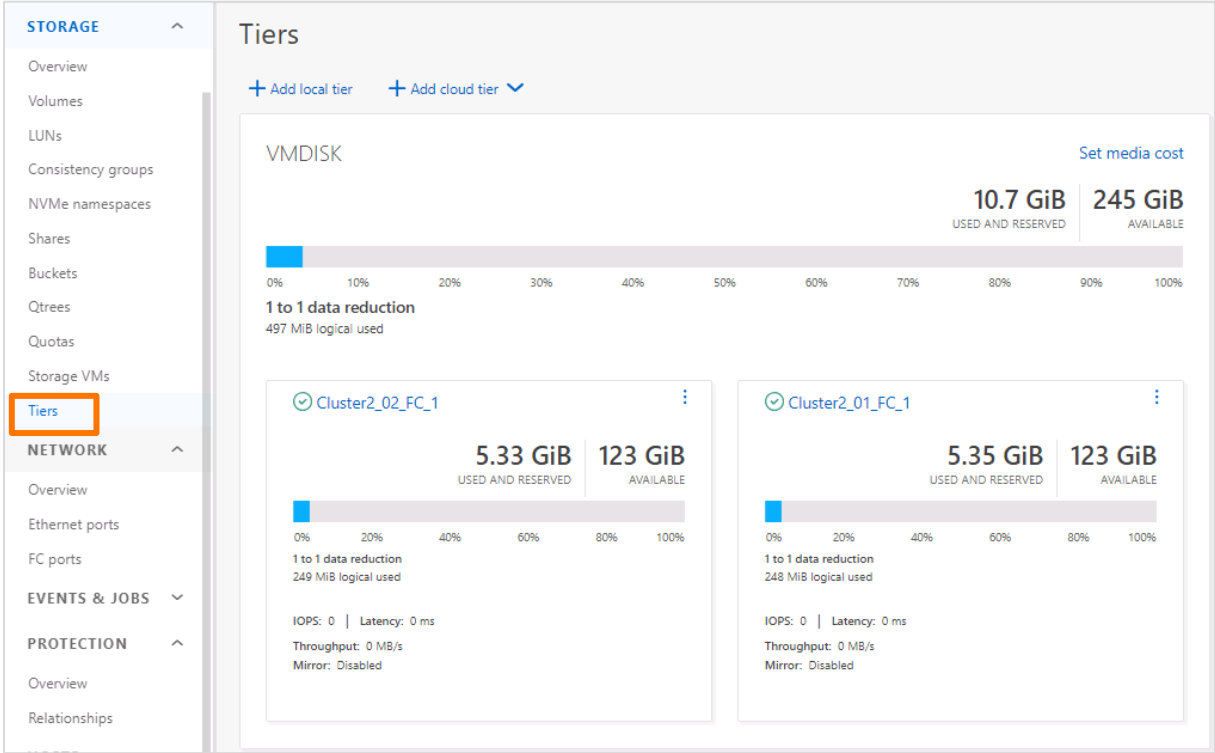
- Verify that space requirements are met
- Create an SVM DR policy
- Create an SVM DR relationship that protects SVM-scoped resources
- Fail over to the SVM on the disaster recovery cluster and verify data access
- Recover and resynchronize the primary source SVM

Task 1: Check Space Requirements

In this task, you examine the primary storage VM svm3 for used space. You also examine the destination aggregates for free space.

Step	Action																																								
1-1	In the System Manager navigation pane for Cluster1, select Storage > Volumes .																																								
1-2	Click the Storage VM column to sort the storage VMs by name. DASHBOARD INSIGHTS STORAGE Overview Volumes LUNs Consistency groups NVMe namespaces Shares Buckets Qtrees Quotas Storage VMs Tiers Volumes + Add : More <table><tr><th>☐</th><th>Name</th><th>Storage VM</th><th>Status</th><th>Capacity</th></tr><tr><td>☒</td><td>c1_svm1_root</td><td>c1_svm1</td><td>Online</td><td>980 KiB used18 MiB available20 MiB</td></tr><tr><td>☒</td><td>c1_svm1_vol1</td><td>c1_svm1</td><td>Online</td><td>242 MiB used730 MiB available1 GiB</td></tr><tr><td>☒</td><td>c1_svm2_root</td><td>c1_svm2</td><td>Online</td><td>972 KiB used16.1 MiB available20 MiB</td></tr><tr><td>☒</td><td>c1_svm2_vol1</td><td>c1_svm2</td><td>Online</td><td>2.24 MiB used971 MiB available1 GiB</td></tr><tr><td>☒</td><td>c1_svm3_root</td><td>c1_svm3</td><td>Online</td><td>1,000 KiB used18 MiB available20 MiB</td></tr><tr><td>☒</td><td>c1_svm3_vol1</td><td>c1_svm3</td><td>Online</td><td>244 MiB used729 MiB available1 GiB</td></tr><tr><td>☒</td><td>c1_svm3_vol2</td><td>c1_svm3</td><td>Online</td><td>1.88 MiB used971 MiB available1 GiB</td></tr></table>	☐	Name	Storage VM	Status	Capacity	☒	c1_svm1_root	c1_svm1	Online	980 KiB used18 MiB available20 MiB	☒	c1_svm1_vol1	c1_svm1	Online	242 MiB used730 MiB available1 GiB	☒	c1_svm2_root	c1_svm2	Online	972 KiB used16.1 MiB available20 MiB	☒	c1_svm2_vol1	c1_svm2	Online	2.24 MiB used971 MiB available1 GiB	☒	c1_svm3_root	c1_svm3	Online	1,000 KiB used18 MiB available20 MiB	☒	c1_svm3_vol1	c1_svm3	Online	244 MiB used729 MiB available1 GiB	☒	c1_svm3_vol2	c1_svm3	Online	1.88 MiB used971 MiB available1 GiB
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1-3	i For each volume, you can view the total capacity, used capacity, and available capacity. To calculate the total aggregate size of the storage that is needed for the disaster recovery cluster, you review the used capacity in each volume in svm3.																																								

Step	Action																																								
1-4	For each volume in svm3, review the used capacity. Volumes + Add ⋮ More <table><thead><tr><th>☐</th><th>Name</th><th>Storage VM</th><th>Status</th><th>Capacity</th></tr></thead><tbody><tr><td>☒</td><td>c1_svm1_root</td><td>c1_svm1</td><td>Online</td><td>980 KiB used 18 MiB available 20 MiB</td></tr><tr><td>☒</td><td>c1_svm1_vol1</td><td>c1_svm1</td><td>Online</td><td>242 MiB used 730 MiB available 1 GiB</td></tr><tr><td>☒</td><td>c1_svm2_root</td><td>c1_svm2</td><td>Online</td><td>972 KiB used 18.1 MiB available 20 MiB</td></tr><tr><td>☒</td><td>c1_svm2_vol1</td><td>c1_svm2</td><td>Online</td><td>2.24 MiB used 971 MiB available 1 GiB</td></tr><tr><td>☒</td><td>c1_svm3_root</td><td>c1_svm3</td><td>Online</td><td>1,000 KiB used 18 MiB available 20 MiB</td></tr><tr><td>☒</td><td>c1_svm3_vol1</td><td>c1_svm3</td><td>Online</td><td>244 MiB used 729 MiB available 1 GiB</td></tr><tr><td>☒</td><td>c1_svm3_vol2</td><td>c1_svm3</td><td>Online</td><td>1.88 MiB used 971 MiB available 1 GiB</td></tr></tbody></table>	☐	Name	Storage VM	Status	Capacity	☒	c1_svm1_root	c1_svm1	Online	980 KiB used 18 MiB available 20 MiB	☒	c1_svm1_vol1	c1_svm1	Online	242 MiB used 730 MiB available 1 GiB	☒	c1_svm2_root	c1_svm2	Online	972 KiB used 18.1 MiB available 20 MiB	☒	c1_svm2_vol1	c1_svm2	Online	2.24 MiB used 971 MiB available 1 GiB	☒	c1_svm3_root	c1_svm3	Online	1,000 KiB used 18 MiB available 20 MiB	☒	c1_svm3_vol1	c1_svm3	Online	244 MiB used 729 MiB available 1 GiB	☒	c1_svm3_vol2	c1_svm3	Online	1.88 MiB used 971 MiB available 1 GiB
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1-5	 To use the CLI to view the used capacity in all the volumes of svm3, you can run the <code>volume show</code> command. <pre>volume show -vserver c1_svm3 -fields used</pre>																																								
1-6	 You must determine how much space is required for the disaster recovery SVM. You must account for the space that the replicated volumes and SVM configuration data require. In a production environment, the SVM configuration data requires at least one non-root aggregate with a minimum of 10GB of free space. The recommended practice for the SVM configuration data is to have at least two non-root aggregates, each with a minimum of 10GB of free space.																																								
1-7	In the System Manager navigation pane for Cluster2, select Storage > Tiers.																																								

Step	Action
1-8	In the Tiers pane, verify that the aggregates can support the size of the volumes in c1_svm3. 
1-9	 To use the CLI to view the used and available capacity in the aggregates on Cluster2, you can run the <code>storage aggregate show</code> command. <pre>cluster2::> storage aggregate show</pre>

Task 2: Create an SVM DR Policy

In this task, you create a custom policy for the SVM DR relationship.

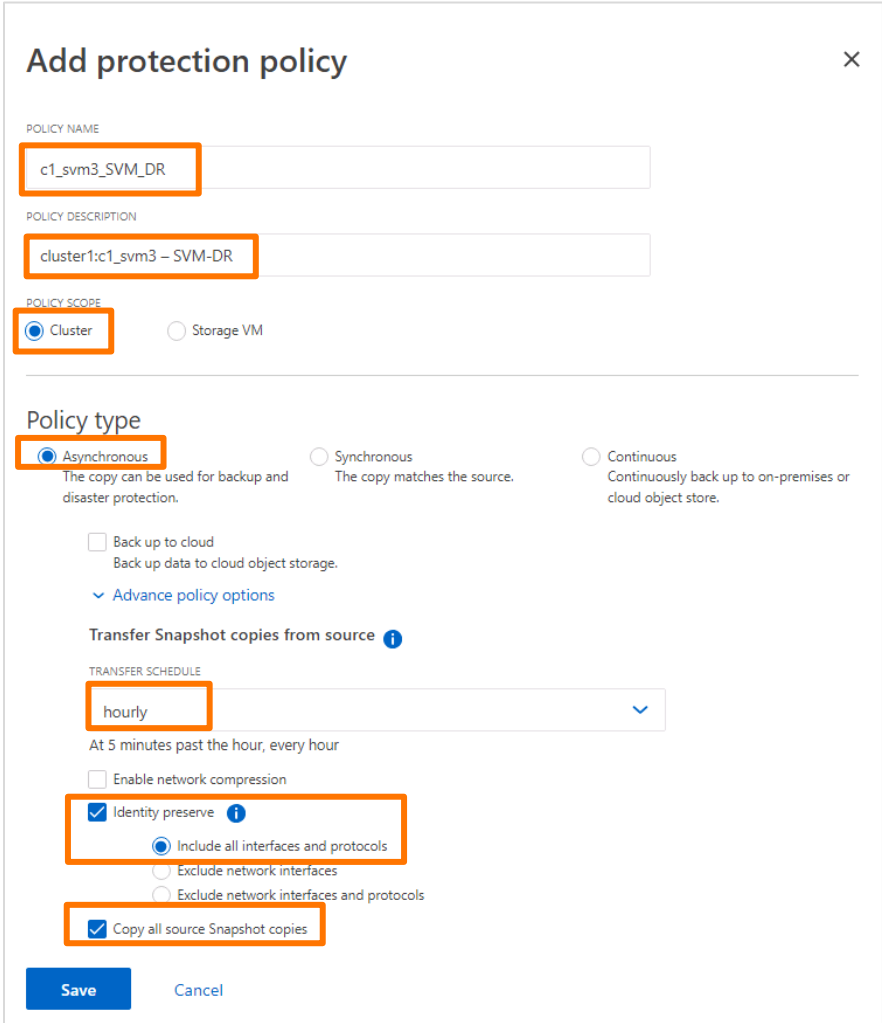
Step	Action
2-1	 Cluster2 is the destination cluster for the protection relationship. You create the custom policy on the destination cluster.

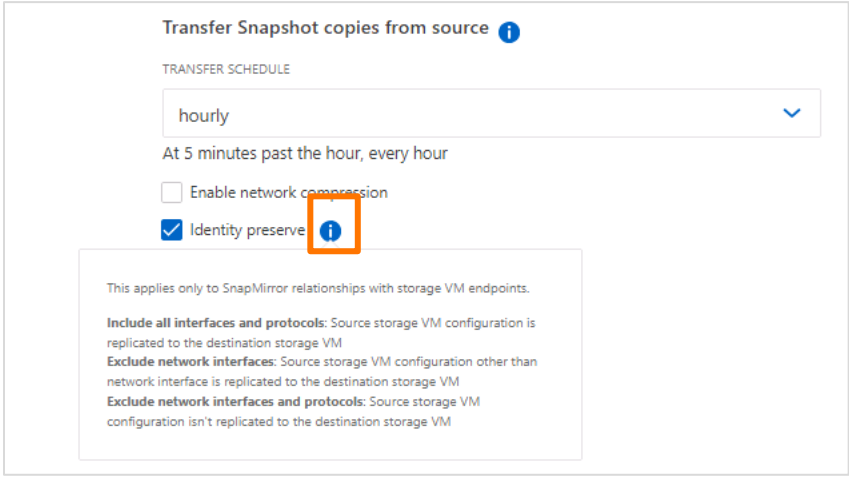
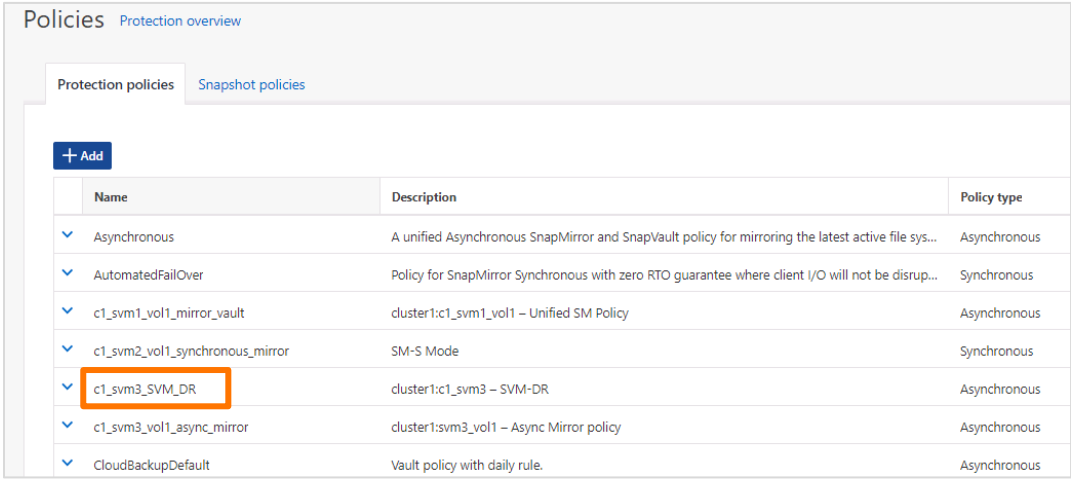
Step	Action
2-2	In the System Manager navigation pane for Cluster2, perform the following steps: - Select Protection > Overview. - In the Local policy settings pane, select Local policy settings and then click the arrow in the upper-right corner of the Protection policies box. - On the Policies > Protection policies tab, click Add to create a policy.

Policies [Protection overview](#)

Protection policies [Snapshot policies](#)

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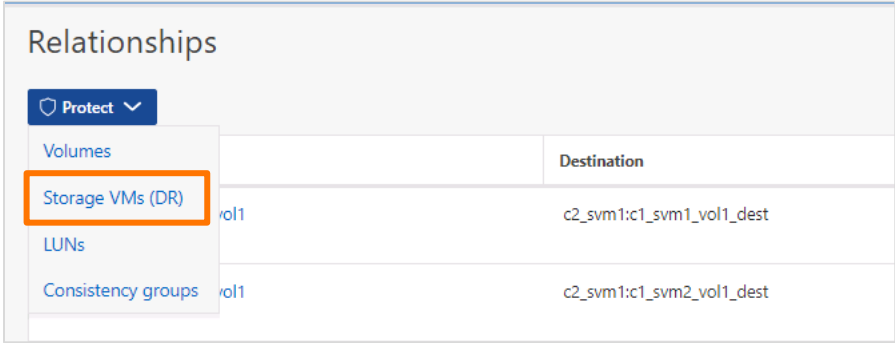
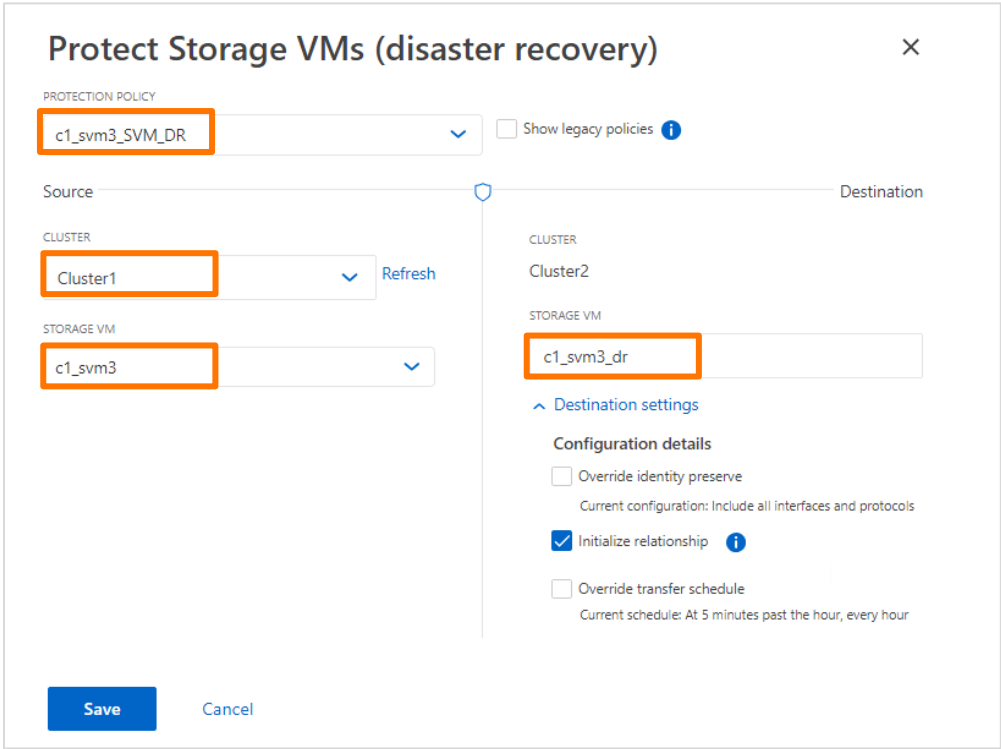
Step	Action
2-3	In the Add protection policy dialog box, specify the following values: - Policy Name: c1_svm3_SVM_DR - Policy Description: cluster1:c1_svm3 – SVM-DR - Policy Scope: Cluster - Policy type: Asynchronous - Transfer Schedule: hourly - Enable network compression: <i>Clear checkbox</i> - Identity preserve: <i>Select checkbox</i> - Copy all source Snapshot copies: <i>Select checkbox</i> 

Step	Action
2-4	In the Add protection policy dialog box, position your cursor over the information icon to the right of the Identity preserve checkbox, and then review the information about the data that is replicated when each suboption is selected. 
2-5	 The source and disaster recovery clusters share a Layer 2 network. To preserve the network configuration that each client uses during a failover event, you use identity preserve mode. In some use cases for identity preserve mode, the administrator might want to modify some network configuration settings after the cutover.
2-6	Click Save .
2-7	Verify that the policy was created successfully. 

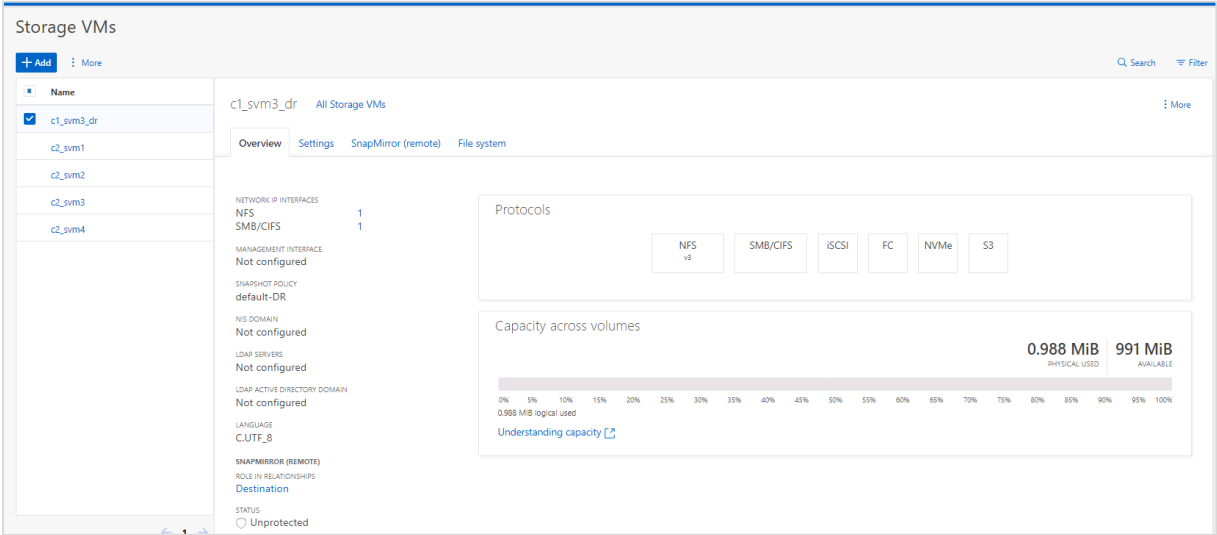
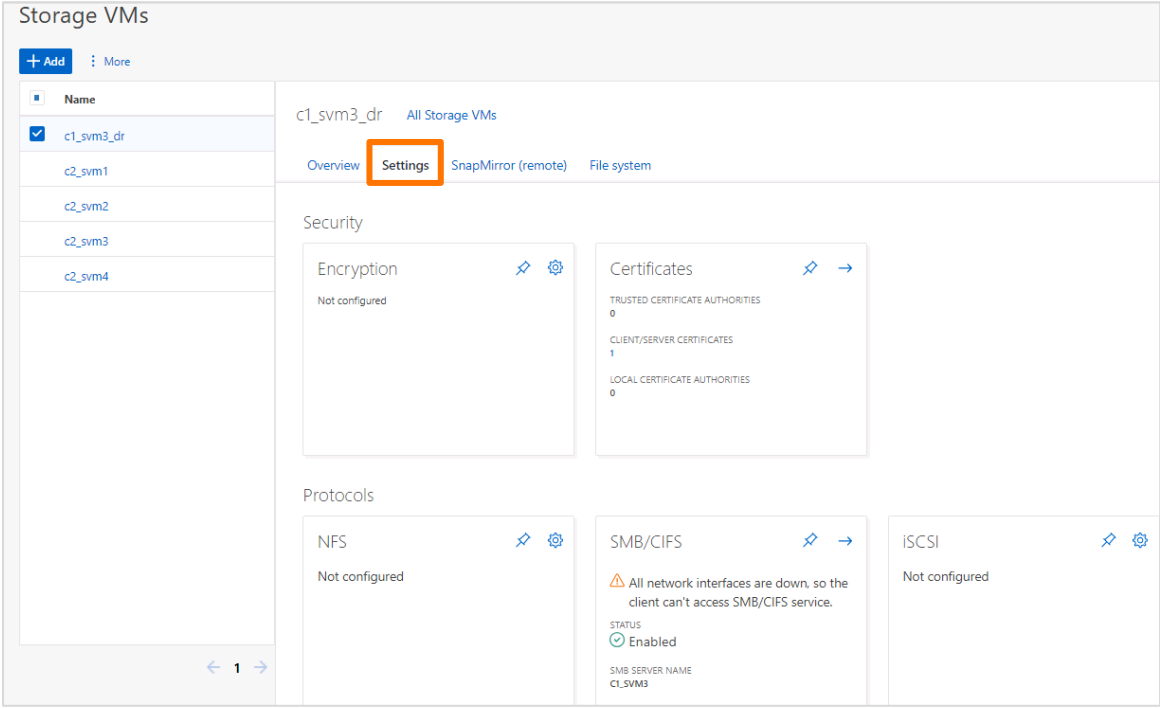
Task 3: Create an SVM DR Relationship

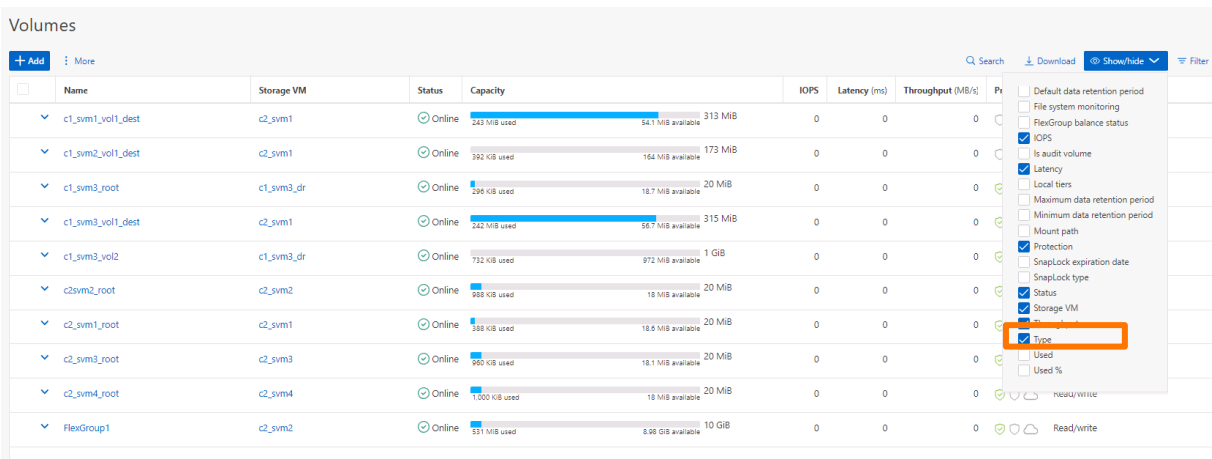
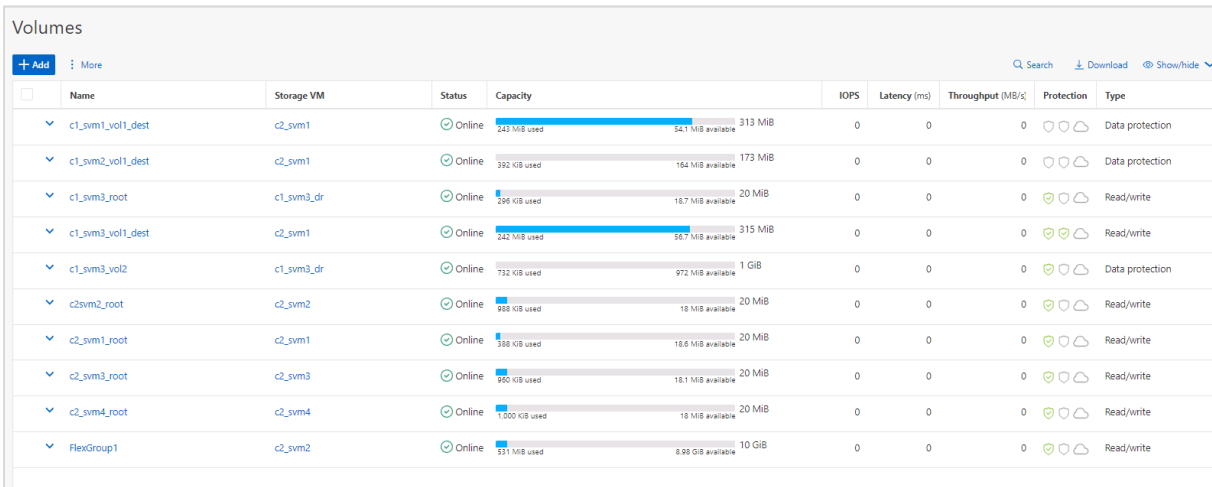
In this task, you create an SVM DR relationship between the primary SVM on Cluster1 and a disaster recovery SVM on Cluster2.

Step	Action
3-1	In the System Manager navigation pane for Cluster2, select Protection > Relationships .

Step	Action						
3-2	Click Protect and then select Storage VMs (DR).  <table border="1"> <thead> <tr> <th>Source</th><th>Destination</th></tr> </thead> <tbody> <tr> <td>vol1</td><td>c2_svm1:c1_svm1_vol1_dest</td></tr> <tr> <td>vol1</td><td>c2_svm1:c1_svm2_vol1_dest</td></tr> </tbody> </table>	Source	Destination	vol1	c2_svm1:c1_svm1_vol1_dest	vol1	c2_svm1:c1_svm2_vol1_dest
Source	Destination						
vol1	c2_svm1:c1_svm1_vol1_dest						
vol1	c2_svm1:c1_svm2_vol1_dest						
3-3	In the Protect Storage VMs (disaster recovery) pane, specify the following values: - Source > Cluster: Cluster1 - Source > Storage VM: c1_svm3 - Destination > Storage VM: c1_svm3_dr 						
3-4	When you select the source storage VM in the previous step, the Protection Policy field appears in which you then specify the Protection Policy as c1_svm3_SVM_DR.						
3-5	Expand Destination Settings and review the information without changing anything.						
3-6	Click Save.						

Step	Action																																																						
3-7	In the Relationships pane, perform the following steps: a. Verify that the new SVM DR relationship is listed. b. Notice how the source and destination are denoted. c. Wait a few minutes to verify that the state is Mirrored. d. Expand the relationship and review the details that are listed. Relationships Protect SearchDownloadShow/Hide <table><tr><th>Source</th><th>Destination</th><th>Protection policy</th><th>Relationship health</th><th>State</th><th>Lag</th></tr><tr><td>c1_svm1:c1_svm1_vol1</td><td>c2_svm1:c1_svm1_vol1_dest</td><td>c1_svm1_vol1_mirror_vault</td><td>Healthy</td><td>Mirrored</td><td>40 minutes, 13 seconds</td></tr><tr><td>c1_svm2:c1_svm2_vol1</td><td>c2_svm1:c1_svm2_vol1_dest</td><td>c1_svm2_vol1_synchronous_mirror</td><td>Healthy</td><td>In sync</td><td>0 second</td></tr><tr><td>c1_svm3:</td><td>c1_svm3_dr:</td><td>c1_svm3_SVM_DR</td><td>Healthy</td><td>Mirrored</td><td>1 minute, 43 seconds</td></tr></table> SOURCE CLUSTER Cluster1 EXPORTED SNAPSHOT COPY vservendr.0.f7b34bdd-f259-11ee-aa51-005056b0191a.2024-04-04_080443 NUMBER OF PROTECTED VOLUMES 1 IDENTITY PRESERVE Include all interfaces and protocols (Derived from policy) DESTINATION CLUSTER Cluster2 POLICY TYPE Asynchronous NUMBER OF PROTECTED CONSISTENCY GROUPS 0 TRANSFER SCHEDULE hourly (Derived from policy) TRANSFER STATUS Success POLICY OPTIONS Mirror only <tr><td>3-8</td><td> After you create the SVM DR relationship, it takes time for the source SVM data to transfer. Wait for the Transfer Status to become Success before proceeding. </td></tr> <tr><td>3-9</td><td> In the System Manager navigation pane, select Storage > Storage VMs. </td></tr> <tr><td>3-10</td><td> In the list, notice that the newly created disaster recovery SVM c1_svm3_dr is in a stopped state and of Subtype dp_destination. Storage VMs Add <table><tr><th>Name</th><th>State</th><th>Subtype</th><th>Configured protocols</th></tr><tr><td>c1_svm3_dr</td><td>stopped</td><td>dp_destination</td><td></td></tr><tr><td>c2_svm1</td><td>running</td><td>default</td><td>NFS, SMB/CIFS</td></tr><tr><td>c2_svm2</td><td>running</td><td>default</td><td>NFS, SMB/CIFS</td></tr><tr><td>c2_svm3</td><td>running</td><td>default</td><td>S3</td></tr><tr><td>c2_svm4</td><td>running</td><td>default</td><td>S3</td></tr></table> </td></tr>	Source	Destination	Protection policy	Relationship health	State	Lag	c1_svm1:c1_svm1_vol1	c2_svm1:c1_svm1_vol1_dest	c1_svm1_vol1_mirror_vault	Healthy	Mirrored	40 minutes, 13 seconds	c1_svm2:c1_svm2_vol1	c2_svm1:c1_svm2_vol1_dest	c1_svm2_vol1_synchronous_mirror	Healthy	In sync	0 second	c1_svm3:	c1_svm3_dr:	c1_svm3_SVM_DR	Healthy	Mirrored	1 minute, 43 seconds	3-8	After you create the SVM DR relationship, it takes time for the source SVM data to transfer. Wait for the Transfer Status to become Success before proceeding.	3-9	In the System Manager navigation pane, select Storage > Storage VMs.	3-10	In the list, notice that the newly created disaster recovery SVM c1_svm3_dr is in a stopped state and of Subtype dp_destination. Storage VMs Add <table><tr><th>Name</th><th>State</th><th>Subtype</th><th>Configured protocols</th></tr><tr><td>c1_svm3_dr</td><td>stopped</td><td>dp_destination</td><td></td></tr><tr><td>c2_svm1</td><td>running</td><td>default</td><td>NFS, SMB/CIFS</td></tr><tr><td>c2_svm2</td><td>running</td><td>default</td><td>NFS, SMB/CIFS</td></tr><tr><td>c2_svm3</td><td>running</td><td>default</td><td>S3</td></tr><tr><td>c2_svm4</td><td>running</td><td>default</td><td>S3</td></tr></table>	Name	State	Subtype	Configured protocols	c1_svm3_dr	stopped	dp_destination		c2_svm1	running	default	NFS, SMB/CIFS	c2_svm2	running	default	NFS, SMB/CIFS	c2_svm3	running	default	S3	c2_svm4	running	default	S3
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Step	Action
3-11	Click c1_svm3_dr to view the details and settings. 
3-12	Click Settings and examine all the settings that are listed for the storage VM. 
3-13	In the System Manager navigation pane, select Storage > Volumes.

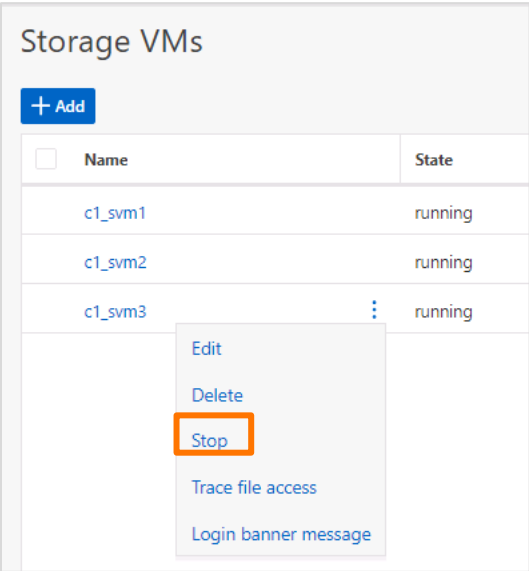
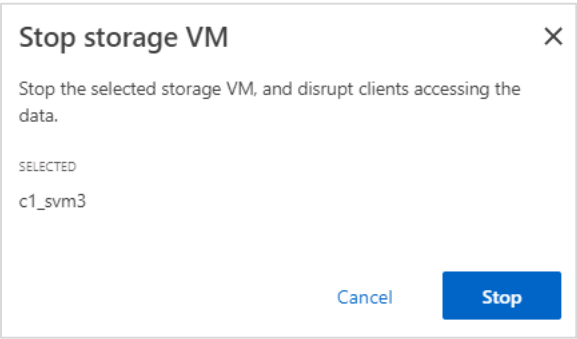
Step	Action
3-14	At the top-right of the Volumes list, click Show/Hide and then select Type to view the Type column. 
3-15	Verify that a new root volume is created automatically for the SVM and that all the source volumes are copied and are of type Data protection. 
3-16	In the System Manager navigation pane, select Storage > Shares.

Step	Action																																																																						
3-17	Examine the shares of the c1_svm3_dr SVM. Shares + Add <table><thead><tr><th></th><th>Name</th><th>Storage VM</th><th>Path</th></tr></thead><tbody><tr><td>▼</td><td>c\$</td><td>c2_svm1</td><td>/</td></tr><tr><td>▼</td><td>c1_svm1_vol1_dest</td><td>c2_svm1</td><td>/c1_svm1_vol1_dest</td></tr><tr><td>▼</td><td>c1_svm3_vol1_dest</td><td>c2_svm1</td><td>/c1_svm3_vol1_dest</td></tr><tr><td>▼</td><td>ipc\$</td><td>c2_svm1</td><td>/</td></tr><tr><td>▼</td><td>c\$</td><td>c2_svm2</td><td>/</td></tr><tr><td>▼</td><td>ipc\$</td><td>c2_svm2</td><td>/</td></tr><tr><td>▼</td><td>c\$</td><td>c1_svm3_dr</td><td>/</td></tr><tr><td>▼</td><td>c1_svm3_share</td><td>c1_svm3_dr</td><td>/vol1</td></tr><tr><td>▼</td><td>ipc\$</td><td>c1_svm3_dr</td><td>/</td></tr></tbody></table> <tr><td>3-18</td><td> In the System Manager navigation pane, select Network > Overview. </td></tr> <tr><td>3-19</td><td> Examine the data LIF of the c1_svm3_dr SVM. Network Interfaces Portsets + Add <table><thead><tr><th>Name</th><th>Status</th><th>Storage VM</th><th>IPspace</th><th>Address</th><th>Current Node</th></tr></thead><tbody><tr><td>c1_svm3_nas_lif1</td><td>!</td><td>c1_svm3_dr</td><td>Default</td><td>192.168.0.62</td><td>cluster2-02</td></tr><tr><td>lif_c2_svm4_921</td><td>✓</td><td>c2_svm4</td><td>Default</td><td>192.168.0.56</td><td>cluster2-01</td></tr><tr><td>lif_c2_svm4_448</td><td>✓</td><td>c2_svm4</td><td>Default</td><td>192.168.0.57</td><td>cluster2-02</td></tr></tbody></table> <tr><td>3-20</td><td> i The data LIF should have the same IP address and network mask as the c1_svm3 SVM on the primary cluster. However, the home port for the LIF might differ from the home port on the source cluster and from what is shown in this guide. The SnapMirror SVM feature makes a best effort at LIF placement and has its own set of rules for LIF home port selection. From the disaster recovery SVM, you cannot access or view the contents of the shares and volumes because you are using identity preserve mode. If you need to access the data in a read-only capacity while the primary cluster actively serves data, you must set identity preserve mode to false. </td></tr></td></tr>		Name	Storage VM	Path	▼	c\$	c2_svm1	/	▼	c1_svm1_vol1_dest	c2_svm1	/c1_svm1_vol1_dest	▼	c1_svm3_vol1_dest	c2_svm1	/c1_svm3_vol1_dest	▼	ipc\$	c2_svm1	/	▼	c\$	c2_svm2	/	▼	ipc\$	c2_svm2	/	▼	c\$	c1_svm3_dr	/	▼	c1_svm3_share	c1_svm3_dr	/vol1	▼	ipc\$	c1_svm3_dr	/	3-18	In the System Manager navigation pane, select Network > Overview.	3-19	Examine the data LIF of the c1_svm3_dr SVM. Network Interfaces Portsets + Add <table><thead><tr><th>Name</th><th>Status</th><th>Storage VM</th><th>IPspace</th><th>Address</th><th>Current Node</th></tr></thead><tbody><tr><td>c1_svm3_nas_lif1</td><td>!</td><td>c1_svm3_dr</td><td>Default</td><td>192.168.0.62</td><td>cluster2-02</td></tr><tr><td>lif_c2_svm4_921</td><td>✓</td><td>c2_svm4</td><td>Default</td><td>192.168.0.56</td><td>cluster2-01</td></tr><tr><td>lif_c2_svm4_448</td><td>✓</td><td>c2_svm4</td><td>Default</td><td>192.168.0.57</td><td>cluster2-02</td></tr></tbody></table> <tr><td>3-20</td><td> i The data LIF should have the same IP address and network mask as the c1_svm3 SVM on the primary cluster. However, the home port for the LIF might differ from the home port on the source cluster and from what is shown in this guide. The SnapMirror SVM feature makes a best effort at LIF placement and has its own set of rules for LIF home port selection. From the disaster recovery SVM, you cannot access or view the contents of the shares and volumes because you are using identity preserve mode. If you need to access the data in a read-only capacity while the primary cluster actively serves data, you must set identity preserve mode to false. </td></tr>	Name	Status	Storage VM	IPspace	Address	Current Node	c1_svm3_nas_lif1	!	c1_svm3_dr	Default	192.168.0.62	cluster2-02	lif_c2_svm4_921	✓	c2_svm4	Default	192.168.0.56	cluster2-01	lif_c2_svm4_448	✓	c2_svm4	Default	192.168.0.57	cluster2-02	3-20	i The data LIF should have the same IP address and network mask as the c1_svm3 SVM on the primary cluster. However, the home port for the LIF might differ from the home port on the source cluster and from what is shown in this guide. The SnapMirror SVM feature makes a best effort at LIF placement and has its own set of rules for LIF home port selection. From the disaster recovery SVM, you cannot access or view the contents of the shares and volumes because you are using identity preserve mode. If you need to access the data in a read-only capacity while the primary cluster actively serves data, you must set identity preserve mode to false.
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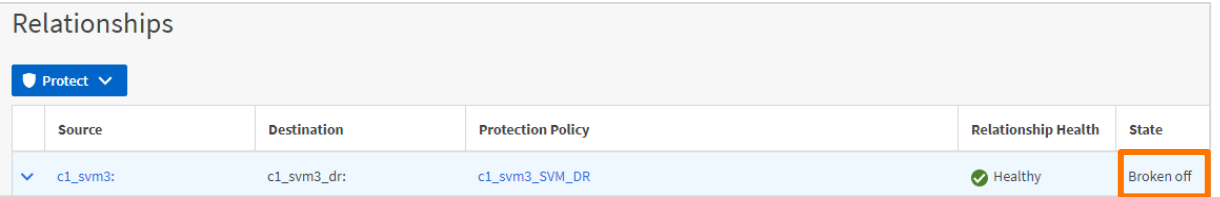
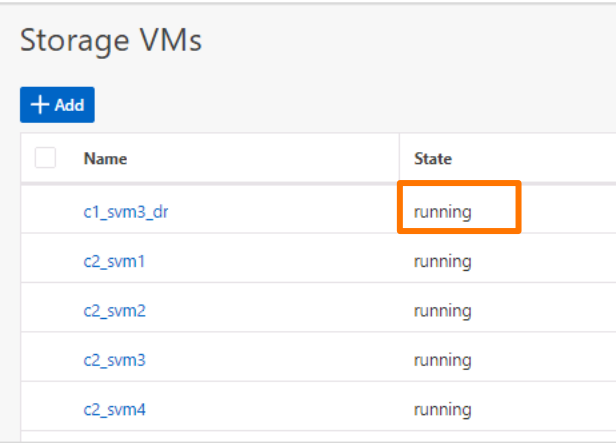
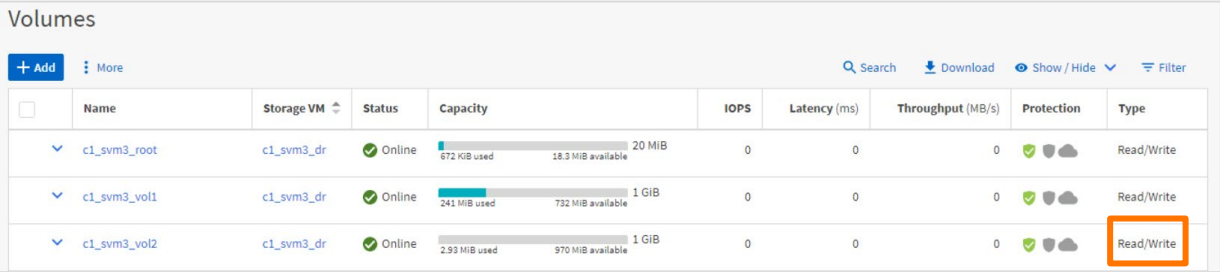
Task 4: Fail Over to the Disaster Recovery SVM

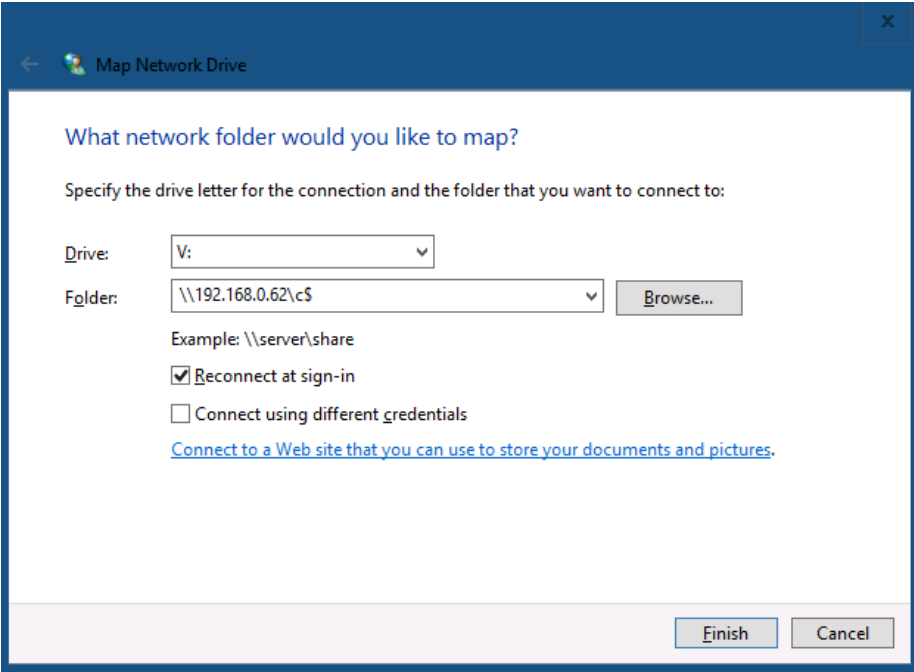
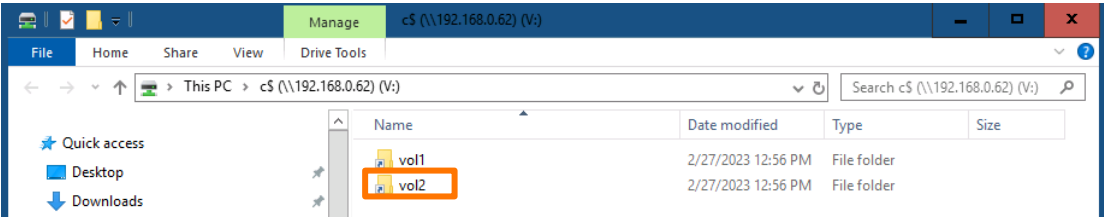
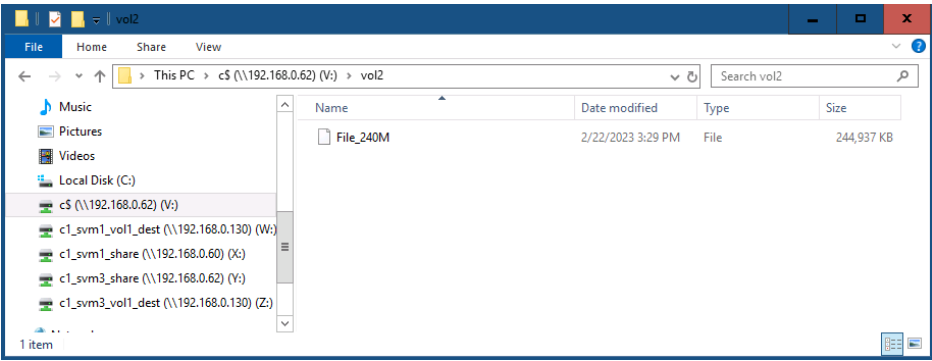
In this task, you recover a system from a disaster on the primary cluster by activating the disaster recovery SVM. This activation involves the following steps:

- 1. Stopping the primary storage VM
- 2. Starting the disaster recovery storage VM
- 3. Verifying the status of the disaster recovery storage VM

Step	Action
4-1	 This exercise uses identity preserve mode for the SnapMirror storage VM relationship. The primary and disaster recovery storage VMs are on the same network subnet. Therefore, you must stop the source storage VM before you activate the destination storage VM.
4-2	In the System Manager navigation pane for Cluster1, select Storage > Storage VMs .
4-3	Click the three vertical dots to the right of c1_svm3 and then select Stop. 
4-4	In the Stop storage VM dialog box, click Stop. 

Step	Action																				
4-5	In the Storage VMs list, verify that the State is stopped. Storage VMs + Add <table><thead><tr><th>☐</th><th>Name</th><th>State</th><th>Subtype</th><th>Configured protocols</th></tr></thead><tbody><tr><td></td><td>c1_svm1</td><td>running</td><td>default</td><td>SMB/CIFS</td></tr><tr><td></td><td>c1_svm2</td><td>running</td><td>default</td><td>SMB/CIFS</td></tr><tr><td></td><td>c1_svm3</td><td>stopped</td><td>default</td><td></td></tr></tbody></table>	☐	Name	State	Subtype	Configured protocols		c1_svm1	running	default	SMB/CIFS		c1_svm2	running	default	SMB/CIFS		c1_svm3	stopped	default	
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4-6	In the System Manager navigation pane for Cluster2, select Protection > Relationships.																				
4-7	In the Source column, click the three vertical dots to the right of c1_svm3: and then select Activate destination storage VM. Relationships Protect <table><thead><tr><th>Source</th><th>Destination</th></tr></thead><tbody><tr><td> c1_svm1:c1_svm1_vol1 </td><td>c2_svm1:c1_svm1_vol1_dest</td></tr><tr><td> c1_svm3:c1_svm3_vol1 </td><td>c2_svm1:c1_svm3_vol1_dest</td></tr><tr><td> c1_svm3: Edit Update Activate destination storage VM c1_svm3_dr:</td><td></td></tr><tr><td> c1_svm2:c1_svm2_vol1 </td><td>c2_svm1:c1_svm2_vol1_dest</td></tr></tbody></table>	Source	Destination	c1_svm1:c1_svm1_vol1	c2_svm1:c1_svm1_vol1_dest	c1_svm3:c1_svm3_vol1	c2_svm1:c1_svm3_vol1_dest	c1_svm3: Edit Update Activate destination storage VM c1_svm3_dr:		c1_svm2:c1_svm2_vol1	c2_svm1:c1_svm2_vol1_dest										
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c1_svm2:c1_svm2_vol1	c2_svm1:c1_svm2_vol1_dest																				
4-8	In the Activate destination storage VM dialog box, select the checkbox and then click Activate. Activate destination storage VM Activates the destination storage VM. The relationship will be broken. The storage VM access will change to read/write mode and will become operational. Source STORAGE VM c1_svm3 Destination STORAGE VM c1_svm3_dr ☒ Activate the destination storage VM and break the relationship. Cancel Activate																				

Step	Action
4-9	 This action breaks the SVM DR relationship, and the c1_svm3_dr SVM starts serving data to clients.
4-10	In the Relationships pane, verify that the SVM DR relationship is in the Broken off state.  The screenshot shows the 'Relationships' pane with a 'Protect' dropdown. A table lists the relationship between 'c1_svm3:' and 'c1_svm3_dr:' with protection policy 'c1_svm3_SVM_DR'. The 'Relationship Health' is 'Healthy' (green checkmark), and the 'State' is 'Broken off' (highlighted with an orange box).
4-11	In the System Manager navigation pane, select Storage > Storage VMs and verify that the storage VM c1_svm3_dr is in a running state.  The screenshot shows the 'Storage VMs' pane with an '+ Add' button. A table lists storage VMs: c1_svm3_dr, c2_svm1, c2_svm2, c2_svm3, and c2_svm4. The 'State' for c1_svm3_dr is 'running' (highlighted with an orange box).
4-12	In the navigation pane, select Storage > Volumes and verify that the type of the volumes has changed from Data protection to Read/Write.  The screenshot shows the 'Volumes' pane with '+ Add' and 'More' buttons, and search/download/show/hide/filter options. A table lists volumes: c1_svm3_root, c1_svm3_vol1, and c1_svm3_vol2. The 'Type' for c1_svm3_vol2 is 'Read/Write' (highlighted with an orange box).
4-13	 Because the primary and disaster recovery storage VMs use the same subnet, the LIFs do not need to be updated. When the storage VM is activated, the clients can access the SMB share. You verify client access in the next steps.
4-14	 To access the volumes from the disaster recovery SVM, you must unmount and remount NFS clients.

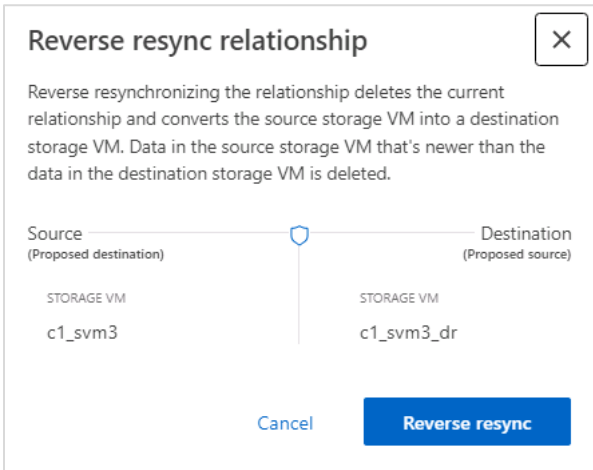
Step	Action
4-15	On the Windows Server desktop, use File Explorer to map a drive letter to the SMB share on c1_svm3_dr on Cluster2: \\192.168.0.62\\c\$. 
4-16	 The c\$ share is the administrative hidden share of the namespace root. You should see folders that represent the junction paths to the other volumes that are involved in the SVM disaster recovery SnapMirror relationship.
4-17	In File Explorer, double-click the vol2 folder, which accesses c1_svm3_vol2. 
4-18	To verify that you can write data to the share, copy File_240M from the CourseFiles folder to the vol2 folder. 

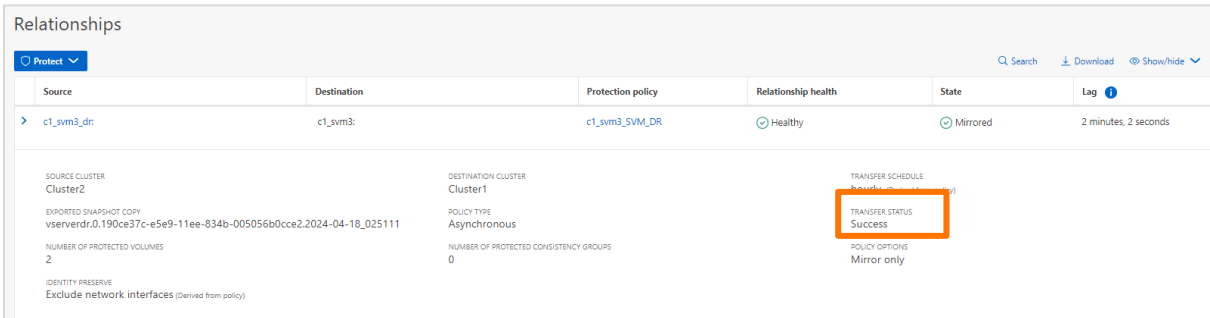
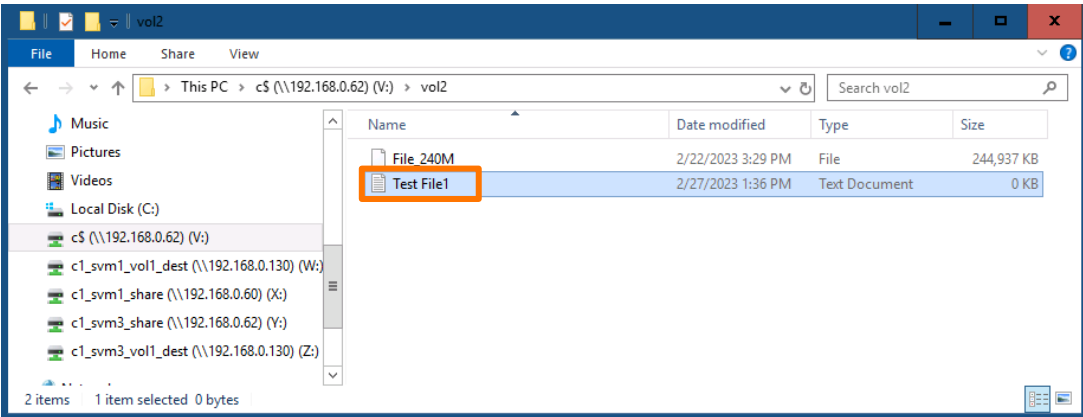
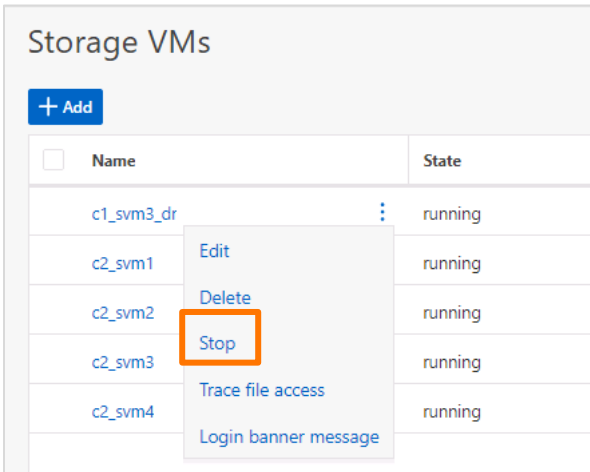
Task 5: Reactivate the Source SVM

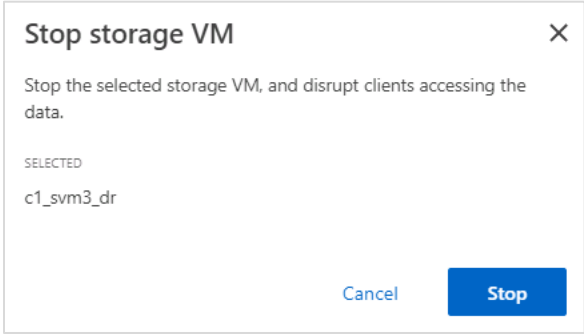
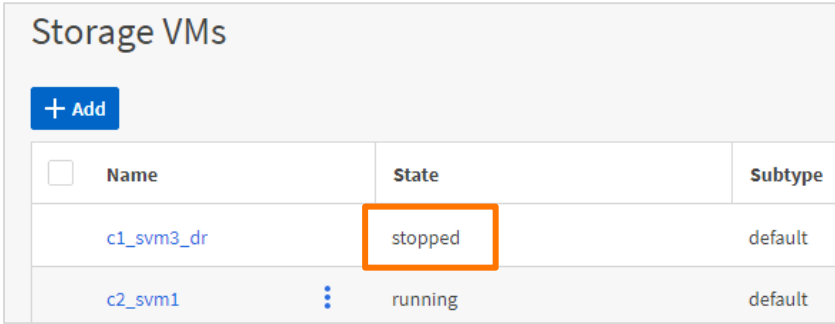
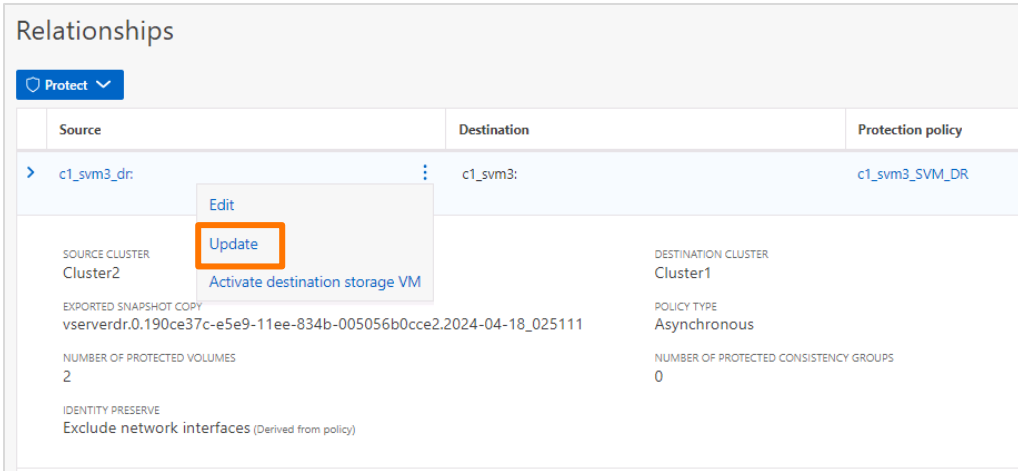
In this task, you activate the source SVM. For these activities, you assume that the source SVM is recoverable. You must plan carefully because a short client outage occurs during switchback. During the reverse relationship, an administrator might also apply SVM configuration changes that are based on new business requirements.

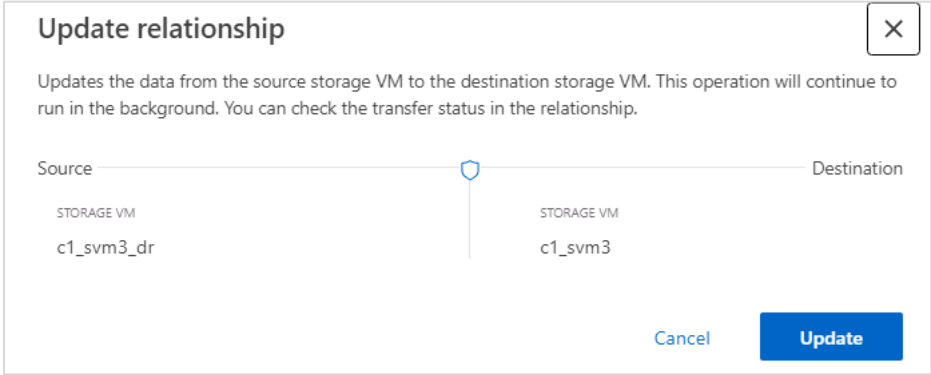
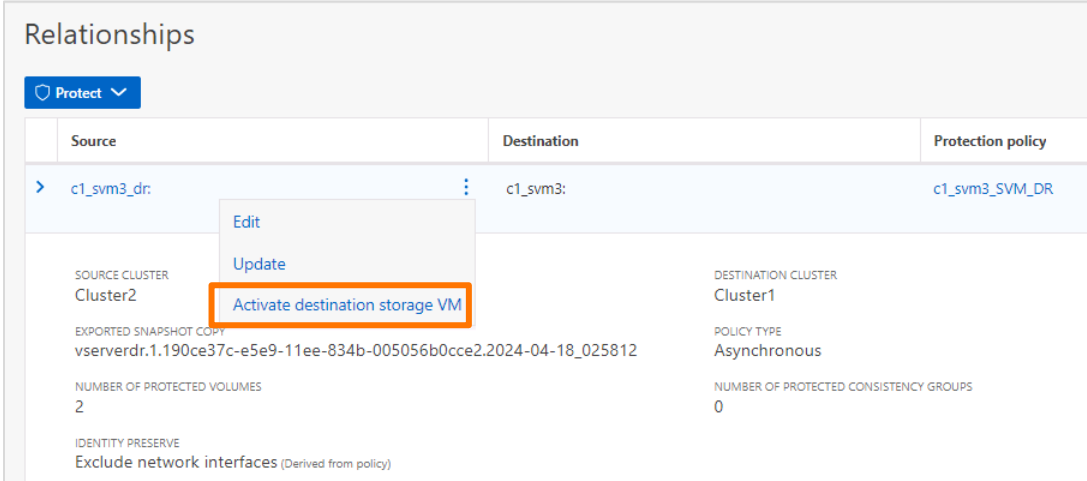
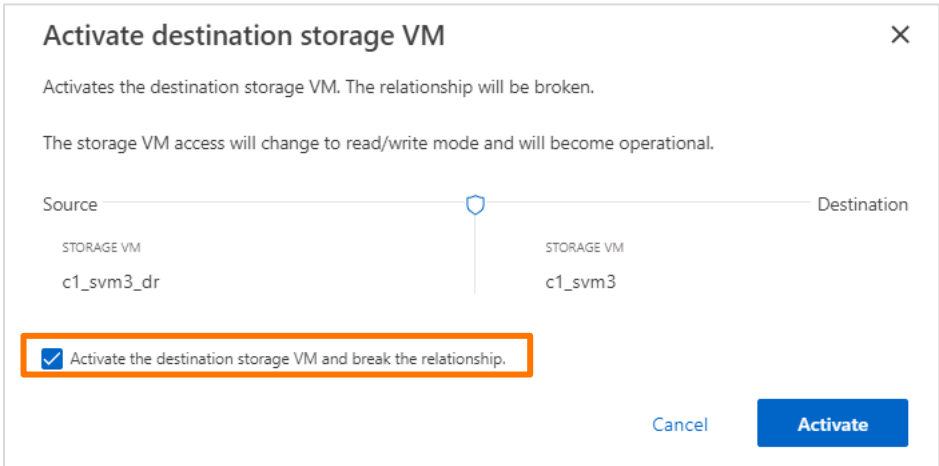
To reactivate the source storage VM, you follow these steps:

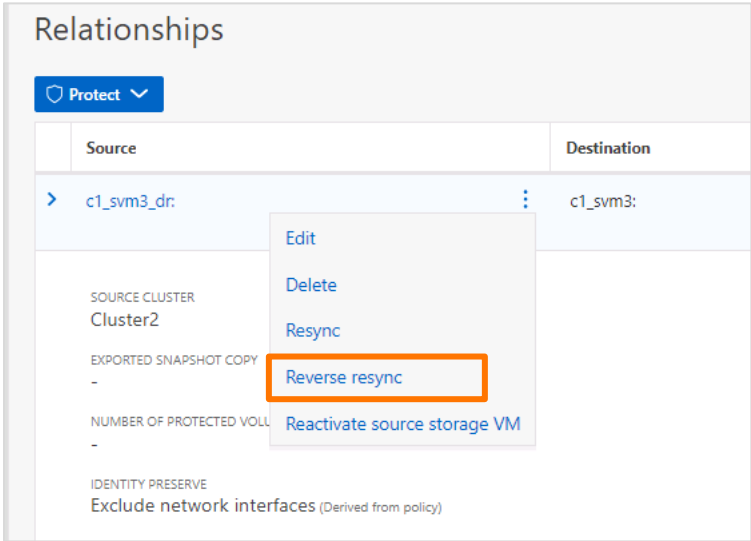
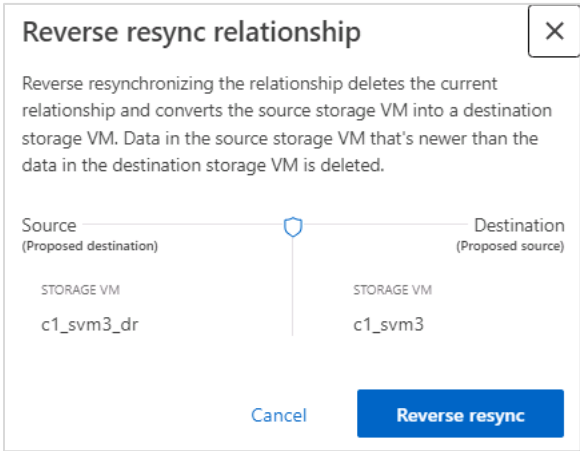
1. Resynchronize the source storage VM from the destination storage VM (reverse resync).
2. Stop the destination storage VM.
3. Update the SnapMirror relationship from the source storage VM.
4. Activate the destination storage VM from the source storage VM.
5. Resynchronize the destination storage VM from the source storage VM (reverse resync).

Step	Action
5-1	 On a live production system, you must verify the presence of various items. For example, on the primary cluster, you must verify the existence of the following: ▪ All required feature licenses and protocols ▪ Any necessary custom schedules ▪ A non-root aggregate with a minimum of 10GB of free space
5-2	In the System Manager navigation pane for Cluster2, select Protection > Relationships .
5-3	In the Source column, click the three vertical dots to the right of svm3 and then select Reverse resync .
5-4	In the Reverse resync relationship dialog box, review the provided information, and then click Reverse resync to resynchronize the data from the destination storage VM to the source storage VM. 
5-5	 You need to wait for the transfer to finish. The SVM DR relationship is then removed from the relationships list.
5-6	In the Relationships pane, verify that the SVM DR relationship was removed.
5-7	In the System Manager navigation pane for Cluster1, select Protection > Relationships .

Step	Action
5-8	Verify that the reversed SVM DR relationship is listed and wait for the Transfer Status to change from Transferring to Success. 
5-9	 You can continue writing to the volumes on the destination storage VM (new source storage VM).
5-10	In File Explorer, create a text file in folder vol2. 
5-11	In the System Manager navigation pane for Cluster2, select Storage > Storage VMs.
5-12	In the Name column, click the three vertical dots to the right of c1_svm3_dr and then select Stop. 

Step	Action
5-13	In the Stop storage VM dialog box, click Stop. 
5-14	In the Storage VMs pane, verify that svm3_dr is in a stopped state. 
5-15	In the System Manager navigation pane for Cluster1, select Protection > Relationships.
5-16	In the Source column, click the three vertical dots to the right of c1_svm3_dr and then select Update. 

Step	Action
5-17	In the Update relationship dialog box, click Update.  The 'Update relationship' dialog box shows a diagram with 'Source' (STORAGE VM c1_svm3_dr) and 'Destination' (STORAGE VM c1_svm3) connected by a line with a shield icon. Below the diagram are 'Cancel' and 'Update' buttons.
5-18	Wait for the Transfer Status to change from Transferring to Success.
5-19	In the Relationships pane, in the Source column, click the three vertical dots to the right of c1_svm3_dr and then select Activate destination storage VM.  The 'Relationships' pane shows a table with columns: Source, Destination, and Protection policy. The first row is highlighted, and a context menu is open over the 'Source' column for 'c1_svm3_dr'. The menu options are 'Edit', 'Update', and 'Activate destination storage VM' (which is highlighted with an orange box). Below the table, details for the source and destination clusters are shown.
5-20	In the Activate destination storage VM dialog box, select the checkbox for “Activate the destination storage VM and break the relationship” and then click Activate to activate the original source SVM c1_svm3.  The 'Activate destination storage VM' dialog box shows a diagram with 'Source' (STORAGE VM c1_svm3_dr) and 'Destination' (STORAGE VM c1_svm3) connected by a line with a shield icon. Below the diagram is a checkbox labeled 'Activate the destination storage VM and break the relationship.' which is checked and highlighted with an orange box. At the bottom are 'Cancel' and 'Activate' buttons.

Step	Action
5-21	In the Relationships pane, in the Source column, click the three vertical dots to the right of c1_svm3_dr and then select Reverse resync.  The screenshot shows the 'Relationships' pane with a 'Protect' dropdown. A table lists relationships with columns 'Source' and 'Destination'. The row for 'c1_svm3_dr' is selected, and a context menu is open with options: Edit, Delete, Resync, Reverse resync (highlighted with an orange box), and Reactivate source storage VM. Below the table, details for 'Cluster2' are shown, including 'EXPORTED SNAPSHOT COPY' and 'NUMBER OF PROTECTED VOLL'.
5-22	In the Reverse resync relationship dialog box, click Reverse resync.  The screenshot shows the 'Reverse resync relationship' dialog box. It contains a warning message: 'Reverse resynchronizing the relationship deletes the current relationship and converts the source storage VM into a destination storage VM. Data in the source storage VM that's newer than the data in the destination storage VM is deleted.' Below the message is a diagram showing 'Source (Proposed destination)' as 'c1_svm3_dr' and 'Destination (Proposed source)' as 'c1_svm3'. At the bottom are 'Cancel' and 'Reverse resync' buttons.
5-23	In the Relationships pane, verify that the SVM DR relationship is removed.
5-24	In the System Manager navigation pane for Cluster2, select Protection > Relationships.

Step	Action																				
5-25	Verify that the original SVM DR relationship is listed and wait for the Transfer Status to change from Transferring to Success. Relationships Protect Search <table><thead><tr><th>Source</th><th>Destination</th><th>Protection policy</th><th>Relationship health</th><th>State</th></tr></thead><tbody><tr><td>c1_svm1:c1_svm1_vol1</td><td>c2_svm1:c1_svm1_vol1_dest</td><td>c1_svm1_vol1_mirror_vault</td><td>Healthy</td><td>Mirrored</td></tr><tr><td>c1_svm3:c1_svm3_vol1</td><td>c2_svm1:c1_svm3_vol1_dest</td><td>c1_svm3_vol1_async_mirror</td><td>Healthy</td><td>Mirrored</td></tr><tr><td>c1_svm3:</td><td>c1_svm3_dr:</td><td>c1_svm3_SVM_DR</td><td>Healthy</td><td>Mirrored</td></tr></tbody></table> SOURCE CLUSTER Cluster1 EXPORTED SNAPSHOT COPY vsvervdr.0.e39987bc-fd22-11ee-abc1-005056b0d71e.2024-04-18_030119 NUMBER OF PROTECTED VOLUMES 2 IDENTITY PRESERVE Exclude network interfaces (Derived from policy) DESTINATION CLUSTER Cluster2 POLICY TYPE Asynchronous NUMBER OF PROTECTED CONSISTENCY GROUPS 0 TRANSFER SCHEDULE hourly (Derived from policy) TRANSFER STATUS Success POLICY OPTIONS Mirror only	Source	Destination	Protection policy	Relationship health	State	c1_svm1:c1_svm1_vol1	c2_svm1:c1_svm1_vol1_dest	c1_svm1_vol1_mirror_vault	Healthy	Mirrored	c1_svm3:c1_svm3_vol1	c2_svm1:c1_svm3_vol1_dest	c1_svm3_vol1_async_mirror	Healthy	Mirrored	c1_svm3:	c1_svm3_dr:	c1_svm3_SVM_DR	Healthy	Mirrored
Source	Destination	Protection policy	Relationship health	State																	
c1_svm1:c1_svm1_vol1	c2_svm1:c1_svm1_vol1_dest	c1_svm1_vol1_mirror_vault	Healthy	Mirrored																	
c1_svm3:c1_svm3_vol1	c2_svm1:c1_svm3_vol1_dest	c1_svm3_vol1_async_mirror	Healthy	Mirrored																	
c1_svm3:	c1_svm3_dr:	c1_svm3_SVM_DR	Healthy	Mirrored																	
5-26	In the navigation pane, select Storage > Storage VMs .																				
5-27	Verify that c1_svm3_dr is in a stopped state, which indicates that data access now occurs from the source SVM. Storage VMs + Add <table><thead><tr><th>Name</th><th>State</th><th>Subtype</th></tr></thead><tbody><tr><td>c1_svm3_dr</td><td>stopped</td><td>dp_destination</td></tr><tr><td>c2_svm1</td><td>running</td><td>default</td></tr></tbody></table>	Name	State	Subtype	c1_svm3_dr	stopped	dp_destination	c2_svm1	running	default											
Name	State	Subtype																			
c1_svm3_dr	stopped	dp_destination																			
c2_svm1	running	default																			
5-28	In the navigation pane, select Storage > Volumes .																				
5-29	Verify that the volumes are again of type Data protection.																				
5-30	On the Windows Server, open File Explorer to verify that you can access the drive that is mapped to c\$ (\\192.168.0.62) and that the files that you added exist in the folder vol2. File Explorer vol2 This PC > c\$ (\\192.168.0.62) (V:) > vol2 File_240M 2/22/2023 3:29 PM File 244,937 KB Test File1 2/27/2023 1:36 PM Text Document 0 KB																				

End of exercise